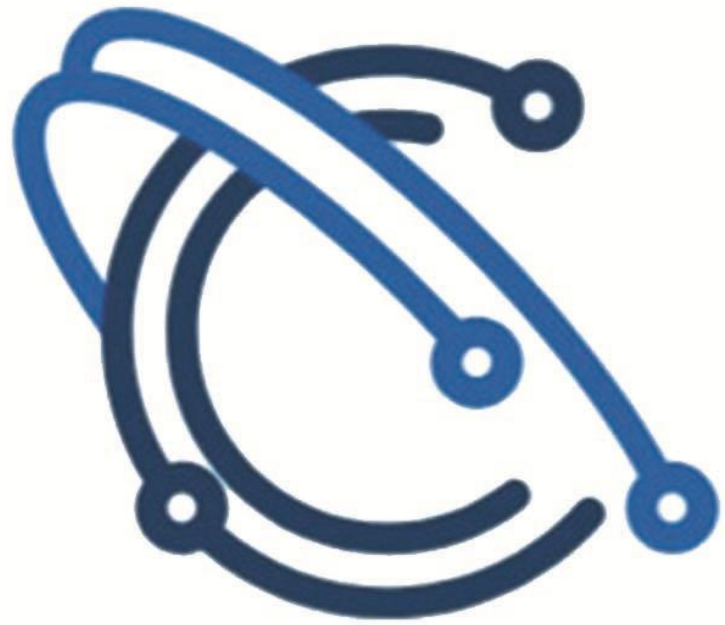




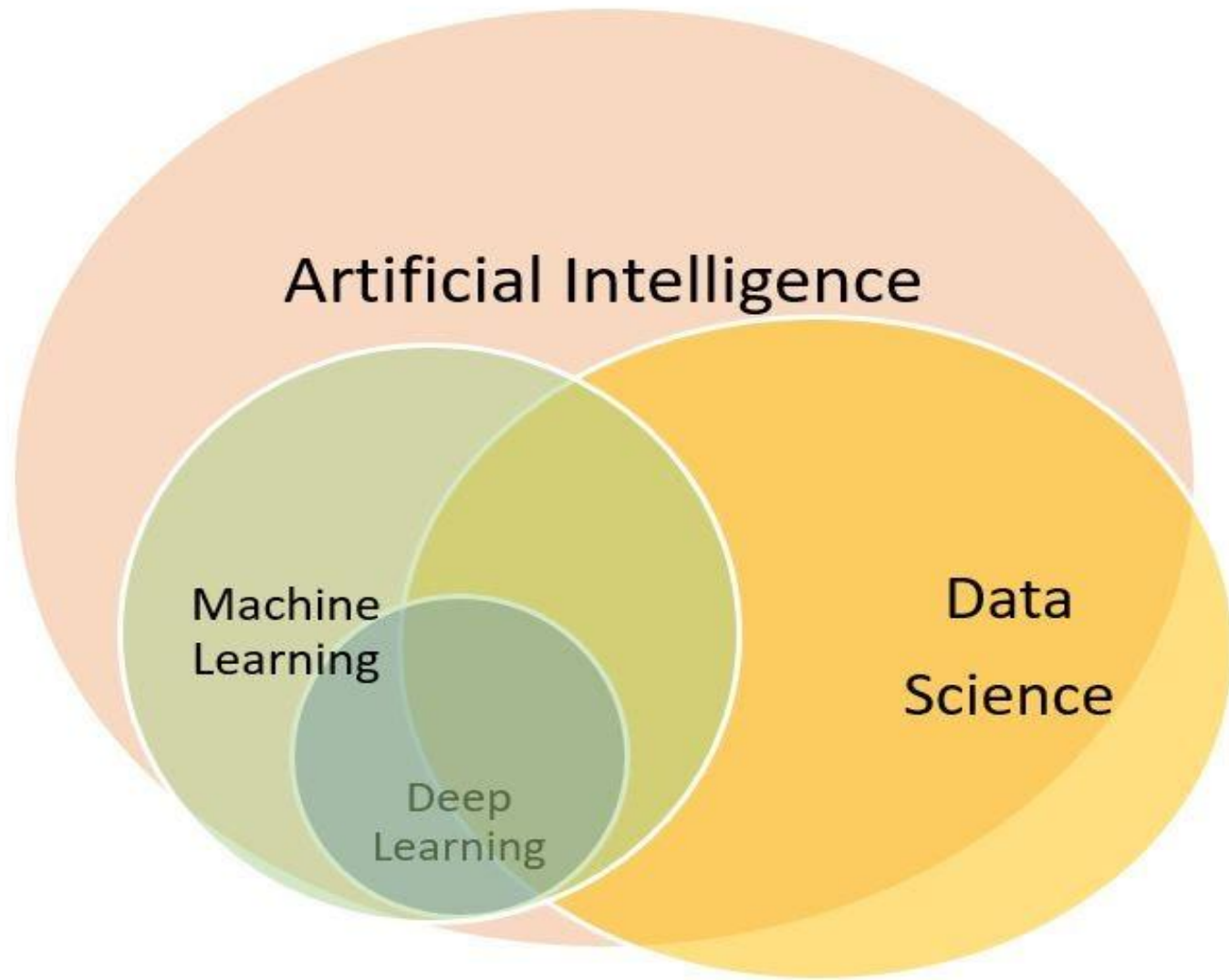
Code Craft

Innovate . Elevate . Inspire

Agenda :

- ✓ Data Science
- ✓ Machine Learning
- ✓ Artificial Intelligence

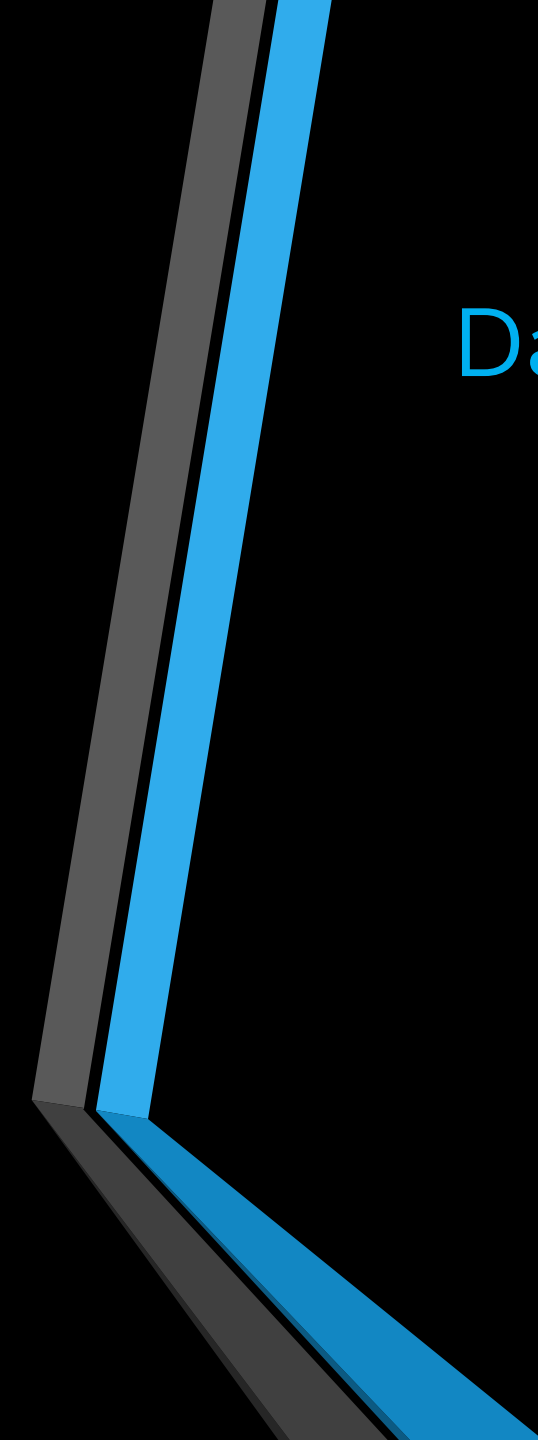
DS / ML / AI



What is Data Science ?



Data Science is the domain of study that deals with vast volumes of data using modern tools and techniques to make business decisions.



Data Analysis :

Descriptive
Diagnostic
Prescriptive
Predictive

Data Science Applications

- **Healthcare**

- Finance
- Retail / E-Commerce
- Logistics
- Gaming
- Search Engines
- Social Media

Predictive Analysis

Image Analysis

Patients Report Analysis

Health Monitoring

Data Science Applications

- Healthcare
- **Finance**
- Retail / E-Commerce
- Logistics
- Gaming
- Search Engines
- Social Media



Stock Market

Data Science Applications

- Healthcare
- Finance
- **Retail / E-Commerce**
- Logistics
- Gaming
- Search Engines
- Social Media

The logo for Flipkart, featuring a yellow shopping cart icon to the left of the word "flipkart" in a stylized, lowercase font, with ".com" in a smaller font to the right.The eBay logo, consisting of the word "ebay" in a lowercase, sans-serif font, with each letter in a different color (e: red, b: blue, a: yellow, y: green).The Amazon logo, featuring the word "amazon" in a lowercase, sans-serif font, with a blue smile arrow underneath it.

Data Science Applications

- Healthcare
- Finance
- Retail / E-Commerce
- **Logistics**
- Gaming
- Search Engines
- Social Media



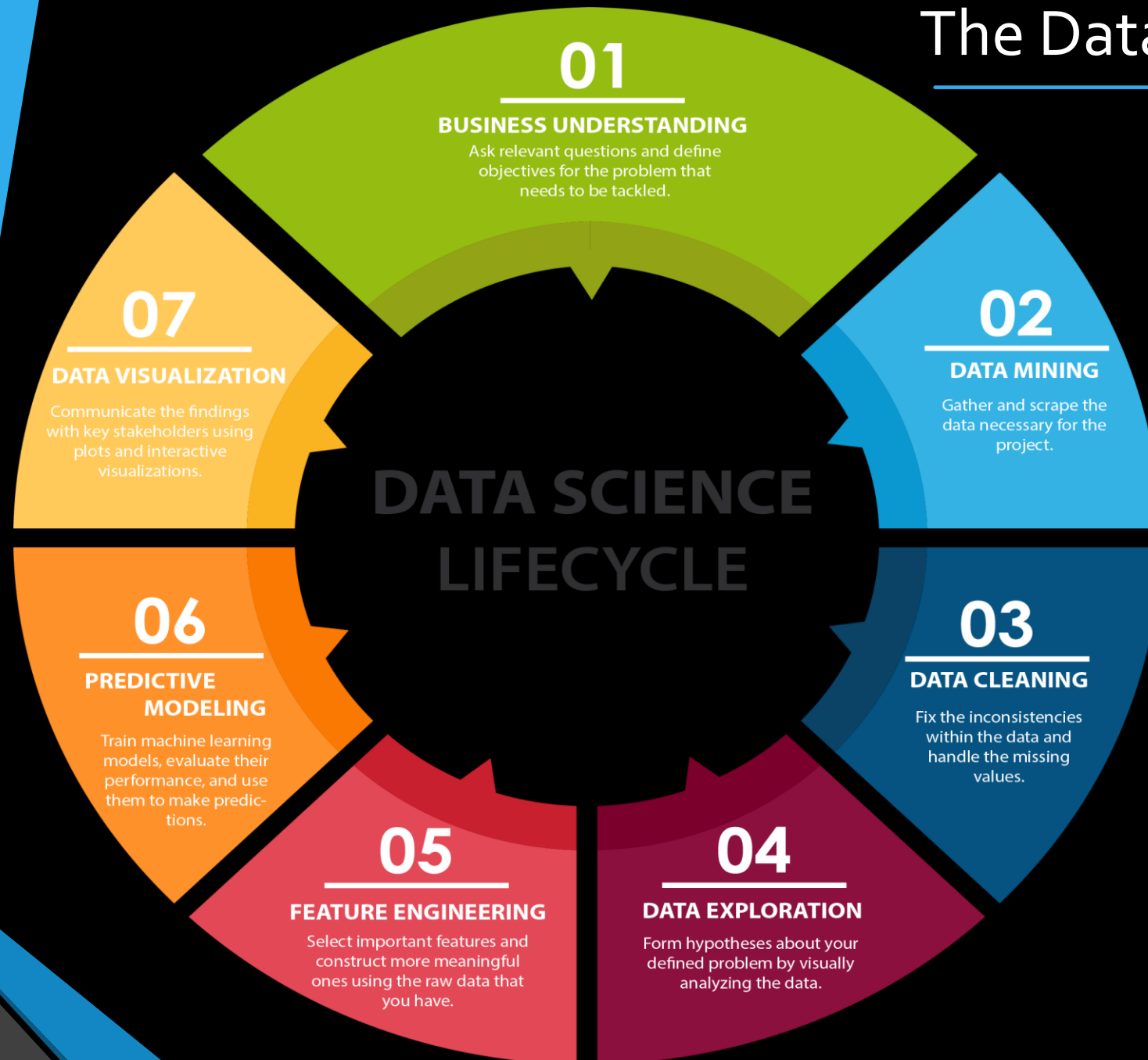
Data Science Applications

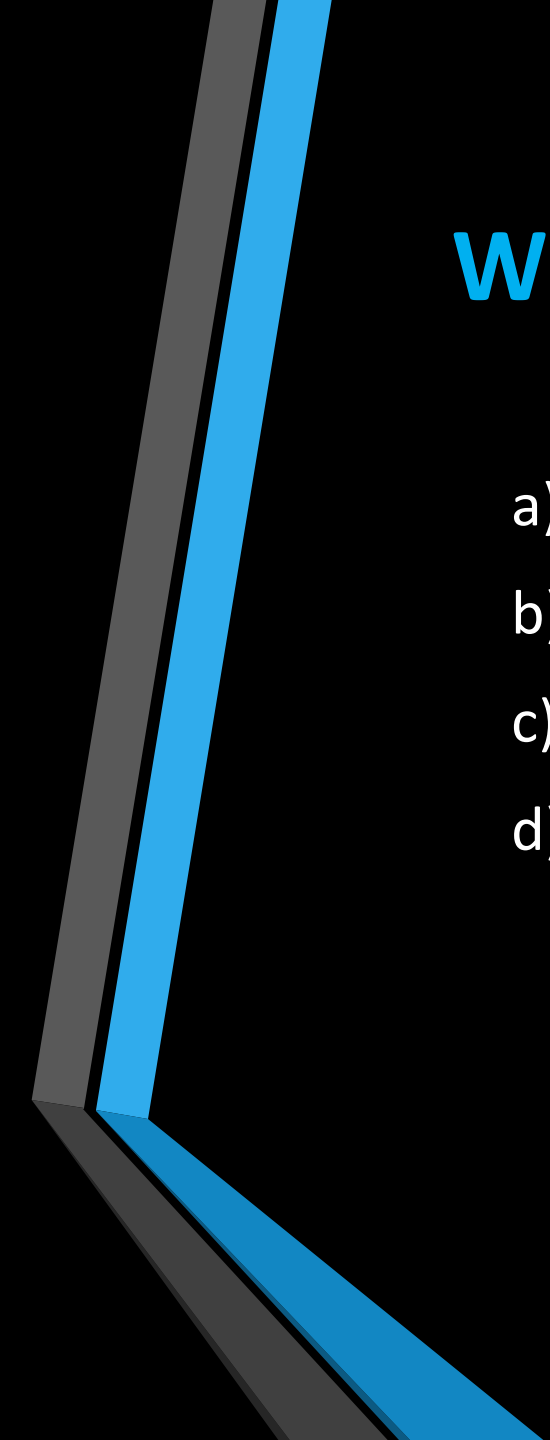
- Healthcare
- Finance
- Retail / E-Commerce
- Logistics
- Gaming
- Search Engines
- Social Media

Many more....



The Data Science Lifecycle





What is the primary goal of data science?

- a) To create visualizations
- b) To extract insights from data
- c) To store large amounts of data
- d) To build websites



Which of the below is not a data science application ?

- Predicting stock prices
- Image recognition
- Generating Random numbers
- Fraud detection

What is Machine Learning

Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and generalize to unseen data and thus perform tasks without explicit instructions.^[1]

What is Machine Learning

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Types of ML

```
graph TD; A[Types of ML] --> B[Supervised]; A --> C[Unsupervised]; A --> D[Semi supervised]; A --> E[Reinforcement]
```

Supervised

Unsupervised

Semi supervised

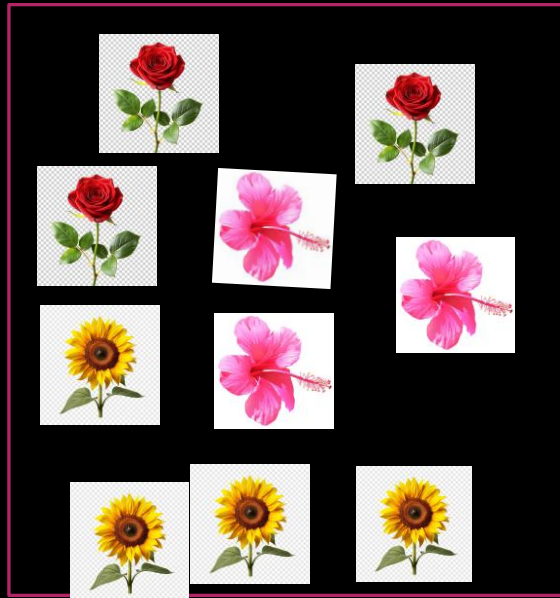
Reinforcement

Supervised ML

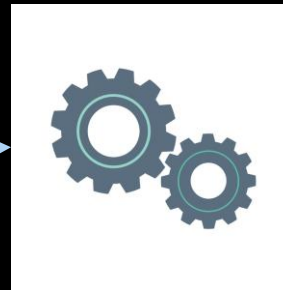
In supervised learning, a machine is trained using a collection of labeled data, where each input is associated with a corresponding desired output. The machine then learns to predict the output for new input data

How Supervised ML works ?

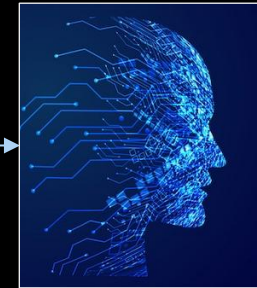
Labeled Data



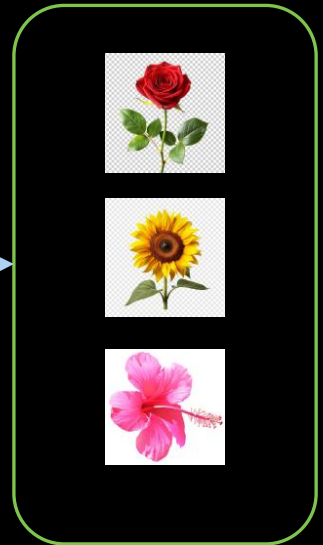
Model Training



Prediction



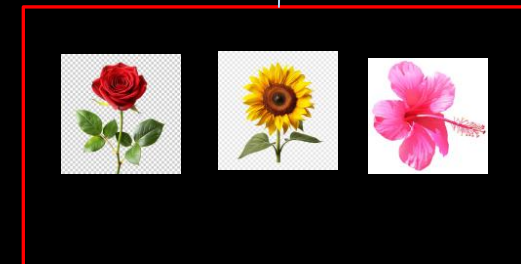
Output



Labels



Test Data



Supervised Learning Algorithms

Regression

- Linear Regression

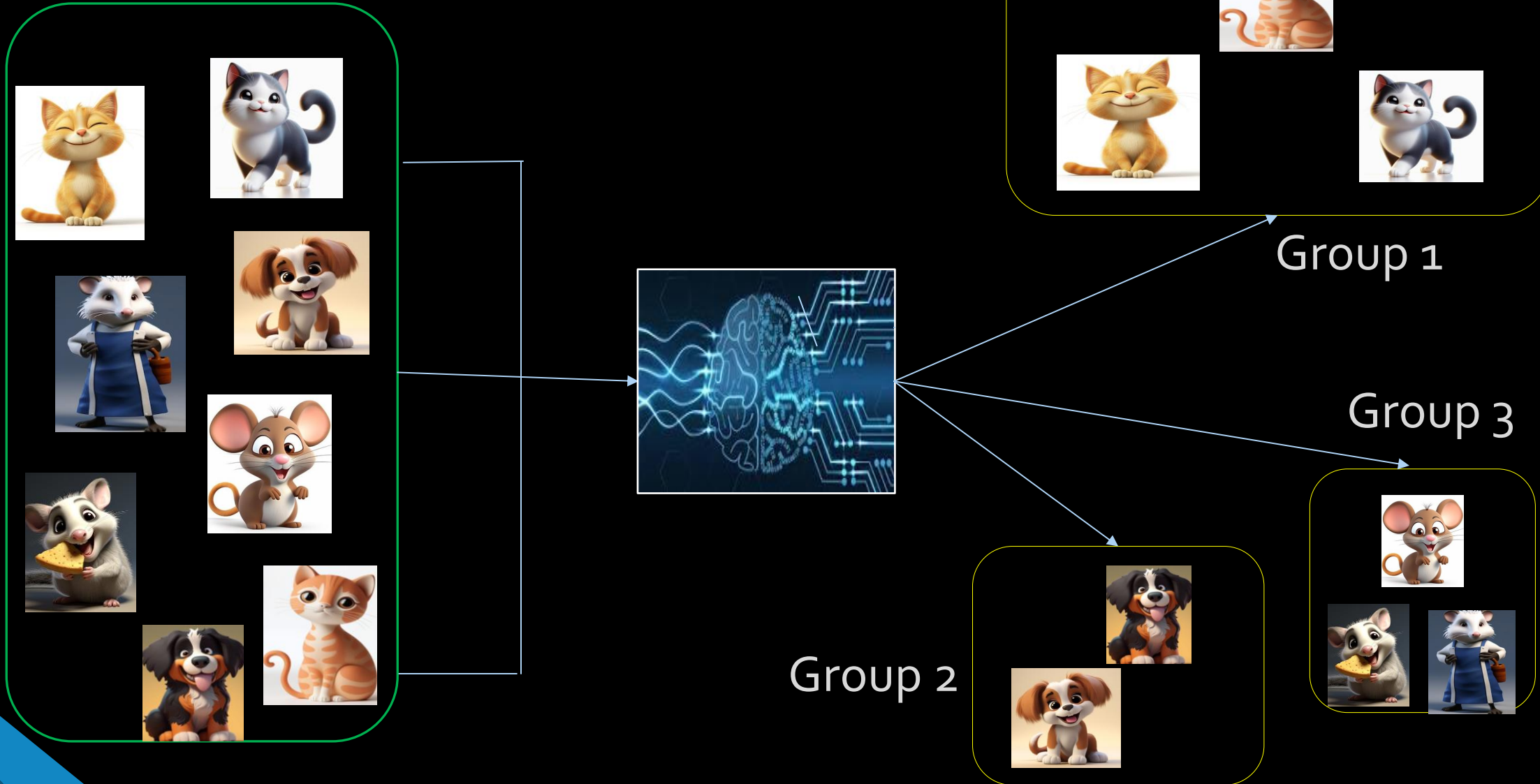
Classification

- Logistic Regression
- Decision Trees
- Random Forests
- Support Vector Machines (SVM)
- Naive Bayes
- K-Nearest Neighbor (KNN)

Unsupervised ML

Unsupervised machine learning analyzes and clusters unlabeled datasets using machine learning algorithms. These algorithms find hidden patterns and data without any human intervention

How Unsupervised ML works ?



Unsupervised Learning Algorithms

- Clustering
- Association Rule Learning
- Dimensionality Reduction

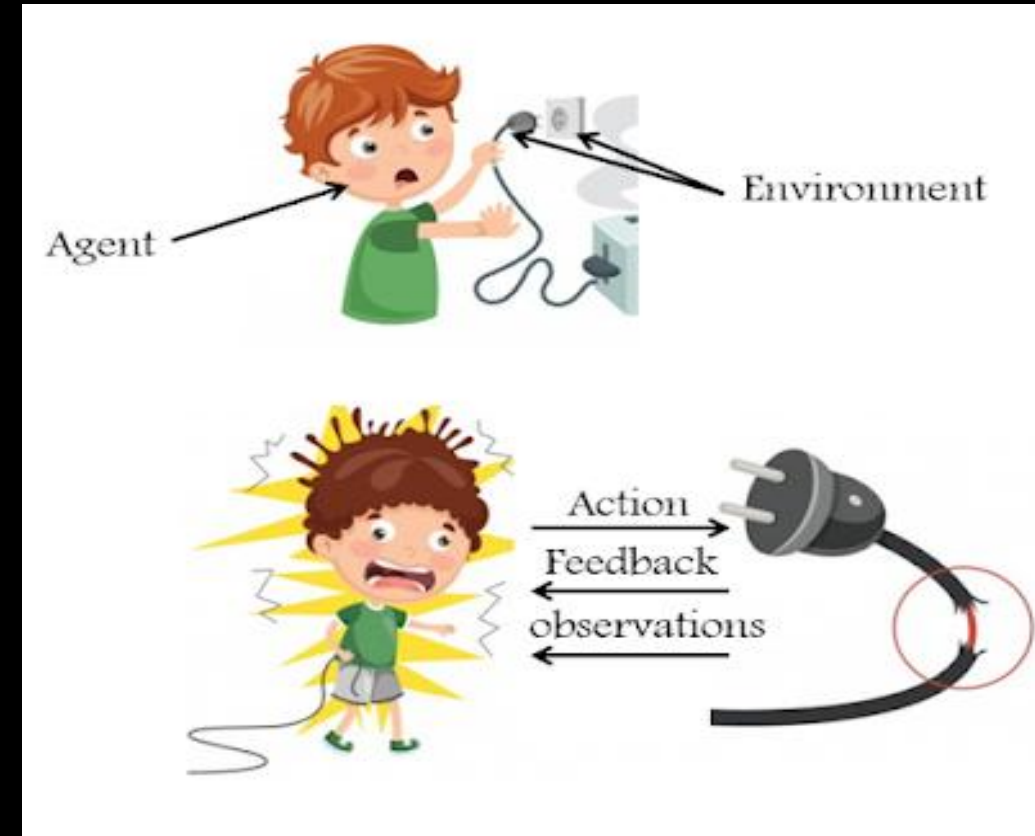
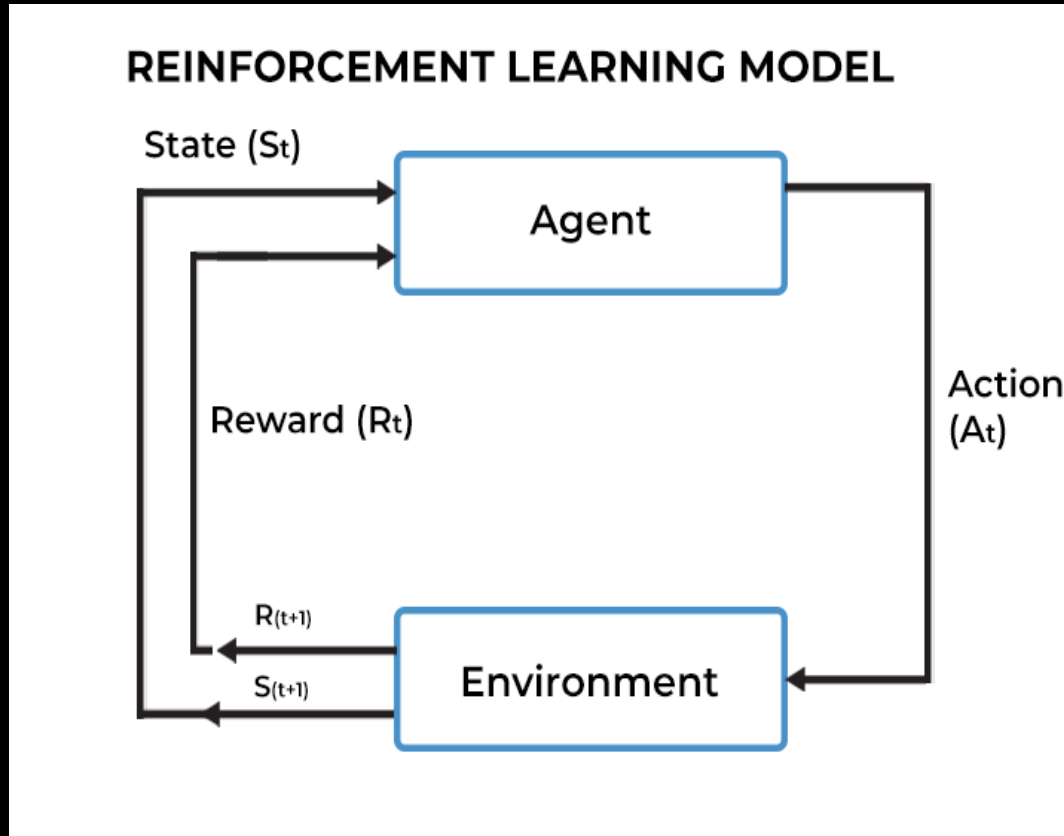
Semi-Supervised Learning

Semi-supervised learning is a type of machine learning that falls in between supervised and unsupervised learning. It is a method that uses a small amount of labeled data and a large amount of unlabeled data to train a model

Reinforcement Learning

Reinforcement Learning is a machine learning approach that relies on feedback, wherein an agent acquires knowledge about how to act within an environment by executing actions and observing the outcomes of those actions.

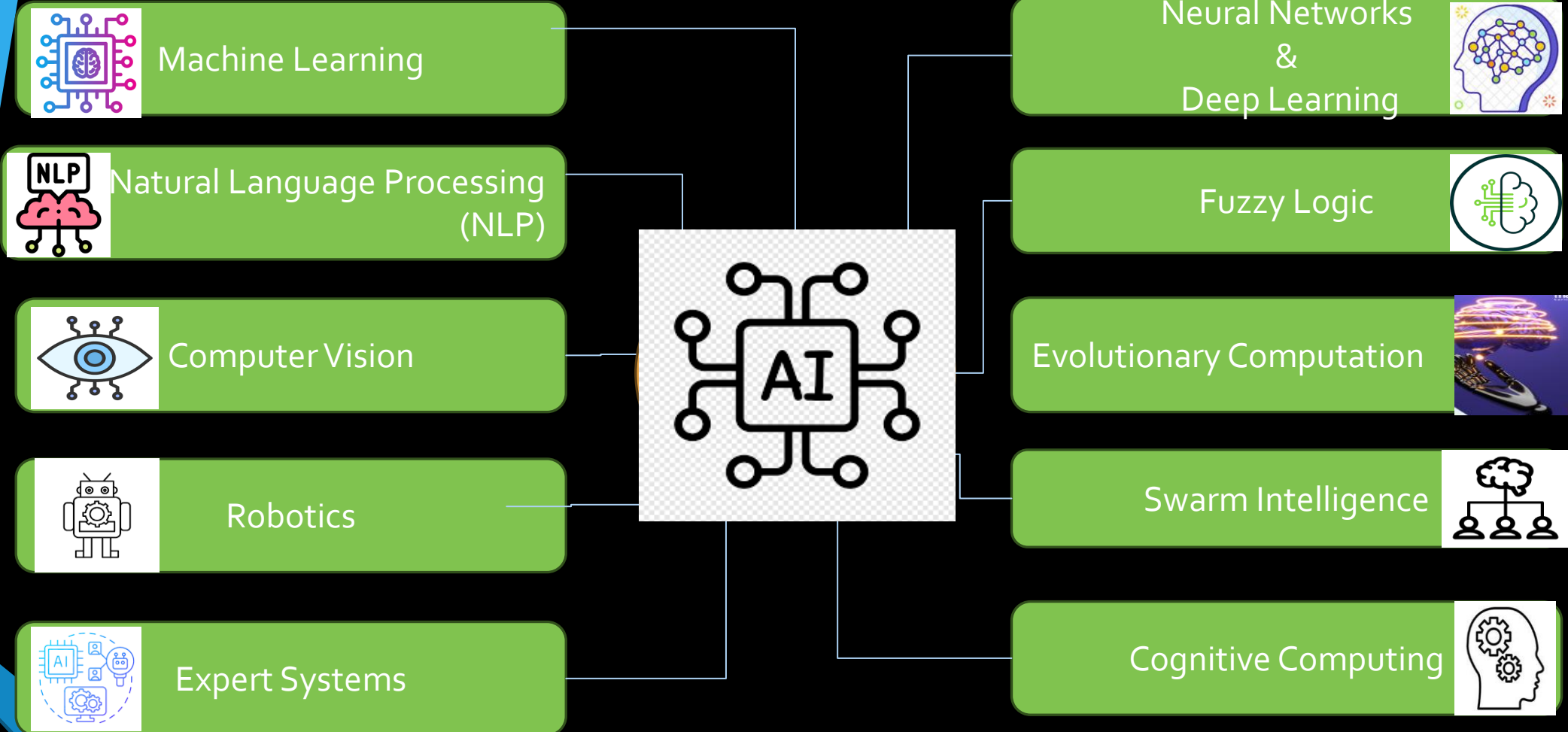
How Reinforcement ML works ?



Artificial Intelligence

Artificial Intelligence is defined as the replication of human intellectual capabilities in machines that are engineered to think, reason, and learn similarly to humans, instead of being directly programmed for specific functions.

Key Components of AI



The Future of Artificial Intelligence

- Advancements in Machine Learning and Deep Learning
- Expansion of Autonomous Systems
- Emergence of General AI
- Advancements in Natural Language Processing and Conversational